



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
28-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-1

Time : 2:45 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **28** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140,

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{2}{\mu_1 d^3}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 15)

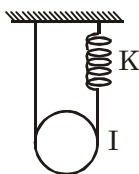
- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : -1 In all other cases

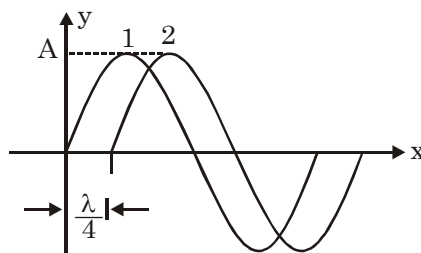
1. The pulley of radius r as shown in figure has a moment of inertia I about its axis and mass m . The time period of vertical oscillation of its centre of mass is, if the spring has spring constant k and the string does not slip over the pulley, is :-



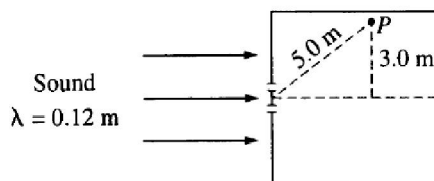
- (A) $2\pi\sqrt{\frac{\left(\frac{I}{r^2} + m\right)}{(2k)}}$ (B) $2\pi\sqrt{\frac{\left(\frac{I}{r^2} - m\right)}{(4k)}}$ (C) $2\pi\sqrt{\frac{\left(\frac{I}{r^2} + m\right)}{(4k)}}$ (D) $2\pi\sqrt{\frac{\left(\frac{I}{r^2} - m\right)}{(2k)}}$
2. A string of 0.1 kg/m and length 0.6 m is fixed at both ends and stretched such that it has a tension of 40 N. The string vibrates in three segments with maximum amplitude of 0.5 cm. Then maximum transverse velocity of a particle which is at a distance $\frac{1}{30}$ m from fixed end is :
 (A) 25π cm/s (B) 25 cm/s (C) 50π cm/s (D) 50 cm/s

Space for Rough Work

3. In the given figure two identical waves each of intensity I_0 ; 1 and 2 are superimposed. The resulting intensity is :



- (A) I_0 (B) $2\sqrt{2}I_0$ (C) $4I_0$ (D) $2I_0$
4. Plane sound waves of wavelength 0.12 m are incident on two narrow slits in a box with non reflecting walls, as shown. At a distance of 5.0 m from the center of the slits, a first-order maximum occurs at point P, which is 3.0 m from the central maximum. The distance between the slits is most nearly :-



- (A) 0.07 m (B) 0.09 m (C) 0.16 m (D) 0.20 m
5. Two thin symmetrical lenses of different nature have equal radii of curvature $R = 20$ cm. The lenses are put close together and immersed in water. The converging focal length of the system is 24 cm. Find the difference between refractive indices of the two lenses. (refractive index for water = $\frac{4}{3}$)

(A) $(\mu_1 - \mu_2) = \frac{1}{24}$

(B) $(\mu_1 - \mu_2) = \frac{1}{20}$

(C) $(\mu_1 - \mu_2) = \frac{5}{9}$

(D) $(\mu_1 - \mu_2) = \frac{4}{3}$

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

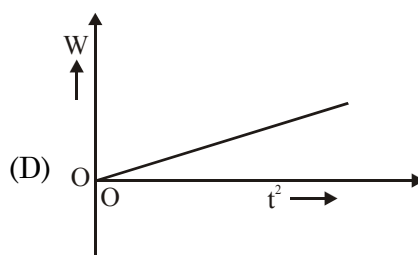
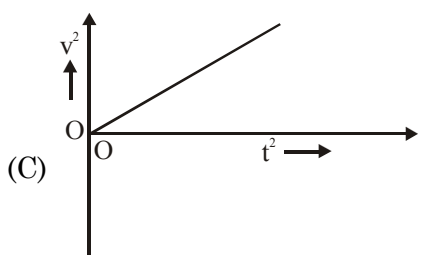
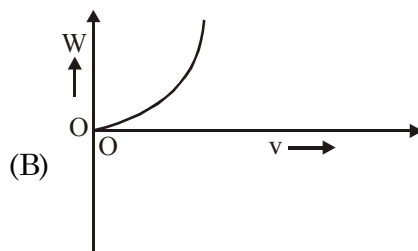
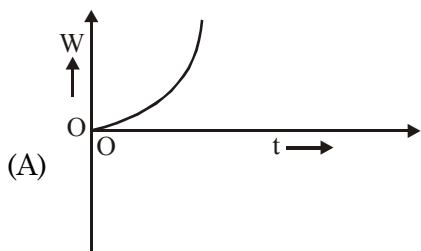
- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

6. Steel blocks A and B, which have equal masses, are at $T_A = 300^\circ\text{C}$ and $T_B = 400^\circ\text{C}$. Block C, with $m_C = 2m_A$ is at $T_C = 350^\circ\text{C}$. Blocks A and B are placed in contact, allowed to come to equilibrium and isolated. Then they are placed in contact with block C. After equilibrium is reached,

(A) $T_A = \sqrt{T_B T_C}$ (B) $T_A = \frac{T_B T_C}{T_B + T_C}$ (C) $\frac{T_A + T_B}{2} = T_C$ (D) $2T_A - T_B = T_C$

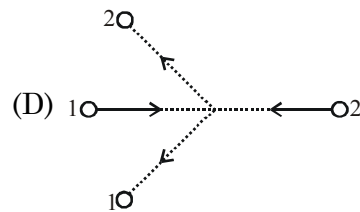
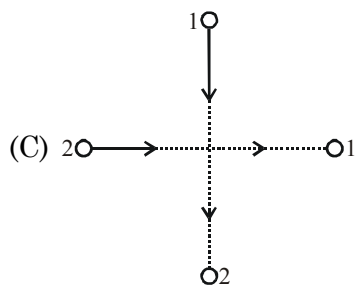
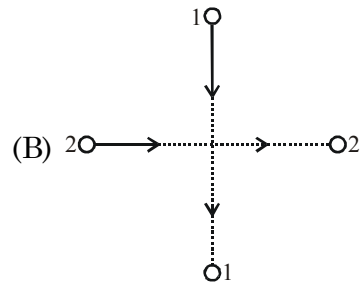
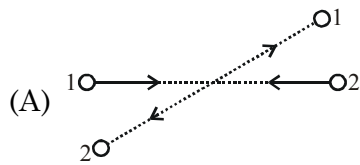
Space for Rough Work

7. A particle initially at rest on a smooth horizontal surface, is acted on by a constant horizontal force at time $t = 0$. Then if W = work done, t = time and v = speed of the particle, the nearly correct graph(s) is/are :-



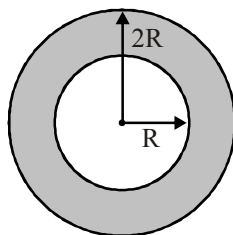
Space for Rough Work

8. Two identical balls (1 and 2) moving with equal speed collide on a frictionless surface. Collision may or may not be elastic. Which of the following diagrams is/are a physically possible situation ?

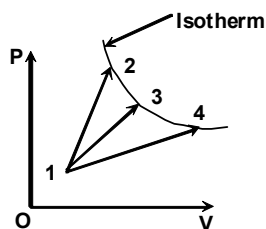


Space for Rough Work

9. Consider a spherical volume which has uniform charge density ρ from $x = R$ to $x = 2R$. There is no charge from $x = 0$ to $x = R$ and from $x = 2R$ to infinity. The electric potential due to this charge :



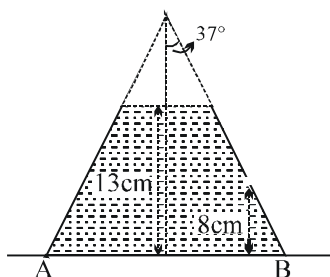
- (A) at $x = 0$ is 0
- (B) at $x = \frac{R}{2}$ is $\frac{3\rho R^2}{2\epsilon_0}$
- (C) at $x = 2R$ is $\frac{7\rho R^2}{6\epsilon_0}$
- (D) at $x = \frac{3R}{2}$ is $\frac{8\rho R^2}{3\epsilon_0}$
10. A gas takes part in three thermal processes in which it is heated from the same initial state to the same final temperature. The process are shown on the P-V diagram by straight lines 1-2, 1-3 and 1-4. Q_1 , Q_2 and Q_3 are amount of heat supplied to the gas in the process $1 \rightarrow 2$, $1 \rightarrow 3$, $1 \rightarrow 4$, respectively.



- (A) $Q_3 > Q_2$
- (B) $Q_1 < Q_2$
- (C) $Q_3 > Q_1$
- (D) $Q_3 > Q_2 > Q_1$

Space for Rough Work

11. A cuboid has sides of 10.4 cm, 3.8 cm and 1.02 cm. Choose the correct statement(s) about the area of the faces and its volume with due regard to significant digits.
- (A) The largest face area is 39.52 cm^2 .
 (B) The smallest face area is 3.9 cm^2 .
 (C) The volume of the cuboid is $4.0 \times 10^1 \text{ cm}^3$.
 (D) The perimeter of the smallest face is 9.64 cm.
12. Curved surface of a vessel has shape of a truncated cone having semi vertex angle 37° . Vessel is full of water (density $\rho = 1000 \text{ kg/m}^3$) upto a height of 13 cm and is placed on a smooth horizontal plane. Upper surface is opened to atmosphere. A hole of 1.5 cm^2 is made on curved wall at a height of 8 cm from bottom as shown in figure. Area of water surface in the vessel is large as compared to the area of hole.



- (A) Initial velocity of efflux is 1 m/sec
 (B) Initial horizontal range of water jet from point B is 6.65 cm
 (C) Horizontal force required to keep the vessel in static equilibrium is 0.15 N
 (D) Horizontal force required to keep the vessel in static equilibrium is 0.12 N.
13. The radionuclide ^{56}Mn has a half-life of 3 hrs and is produced in a cyclotron by bombarding a manganese target with deuterons. The target contains only the stable manganese isotope ^{55}Mn . If the bombardment lasts much longer than 3 hrs, the activity of ^{56}Mn reaches a steady state value of 6.93×10^{10} decay per sec. Assume that all the incident deuterons cause nuclear reaction.
- (A) Apart from ^{56}Mn , a proton is also produced in the nuclear reaction
 (B) The deuterons are incident on the sample at a rate of 6.93×10^{10} per sec
 (C) The steady state mass of ^{56}Mn is approx. 100 ng. (here ng is nano gram)
 (D) Apart from ^{56}Mn , a neutron is also produced in the nuclear reaction

Space for Rough Work

SECTION-II : (Maximum Marks : 15)

MATCHING LIST TYPE QUESTION (ANSWER AS TRIPLE DIGIT INTEGER)

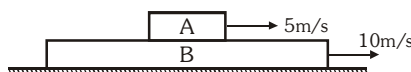
- This section contains **FIVE** questions.
- The answer to each question is a **TRIPLE DIGIT INTEGER** ranging from 111 to 333, both inclusive
- This section contains four questions, each having two matching lists. Choices for the correct combination of elements form List-I (P), (Q), (R) and List-II (1), (2), (3)
- Given matching type questions in which list-I (PQR) match with list-II (1, 2, 3) and represent your answer as triplet digit integer for example correct match for $P \rightarrow 2$; $Q \rightarrow 3$; $R \rightarrow 1$ Then you have fill the answer in section-II as 231.

	(+)	(-)
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Each matching in list-II may be/may not be correct/can be used more than one times for matching in list-I. For example your answer can be 223, 333, 133 etc...

- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

- Block A of mass 4 kg is placed on the rough surface of a long block B of mass 6 kg. Block B is placed on smooth horizontal surface. Blocks are given the velocities as shown in figure.



List-I	List-II
(P) Common velocity (in m/s) of blocks when sliding between A and B stops.	(1) 108
(Q) Velocity (in m/s) of centre of mass at shown moment.	(2) 80
(R) Magnitude of total work done (in joule) by friction on B.	(3) 8

Space for Rough Work

2. A particle of mass m is projected with velocity v and at an angle θ with horizontal then match the following list.

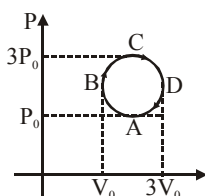
List-I

- (P) Change in momentum between point of projection and point at which it strikes on horizontal surface
(Q) Radius of curvature at highest point
(R) Minimum momentum is

List-II

- (1) $mv \cos\theta$
(2) $2mv \sin\theta$
(3) $\frac{u^2 \cos^2 \theta}{g}$

3. A monatomic ideal gas is taken through a cyclic process ABCDA. ABCDA is circular path. If work done by the gas is denoted by W then match the List-I with List-II.



List-I

- (P) AB
(Q) AC
(R) CD

List-II

- (1) $W > 0$
(2) $W < 0$
(3) $W = 0$

Space for Rough Work

4. Match the following:

List I

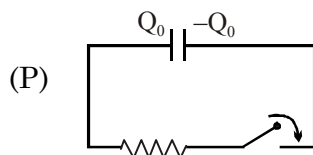
- (P) Thermal resistance
(Q) Stefan's constant
(R) Wien's constant

List II

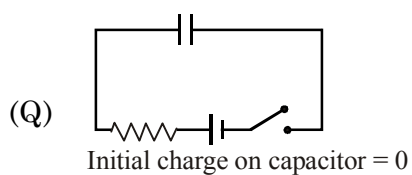
- (1) $[MT^{-3}\theta^{-4}]$
(2) $[M^{-1}L^{-2}T^3\theta]$
(3) $[L\theta]$

5. **List-I**

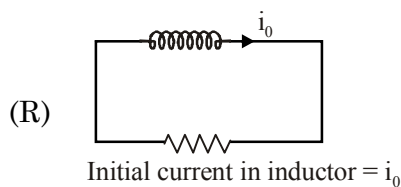
List-II



- (1) Time constant is the time in which current drops to 37% of its initial value.



- (2) Time constant is the time in which current increases to 63% of its steady state value.



- (3) Time constant is the time in which current increases to 37% of its steady state value.

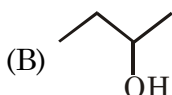
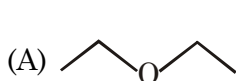
Space for Rough Work

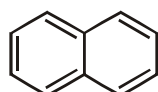
PART-2 : CHEMISTRY

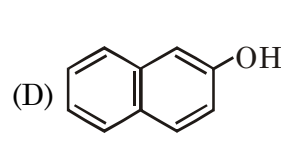
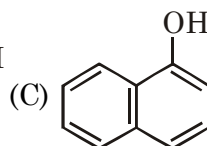
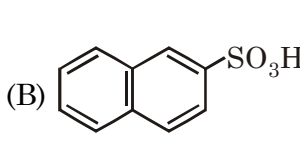
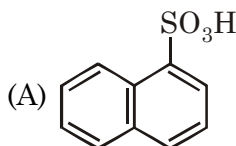
SECTION-I(i) : (Maximum Marks : 15)

- This section contains **FIVE** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
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- 200 ml of 0.005M AgNO_3 reacts with 300 ml of 0.01M KCl . If K_{sp} of AgCl is 1.8×10^{-10} , then maximum concentration of Ag^+ in the mixture is -
 (A) $6 \times 10^{-3}\text{M}$ (B) $4 \times 10^{-3}\text{M}$ (C) $2 \times 10^{-3}\text{M}$ (D) $4.5 \times 10^{-8}\text{M}$
- For particles having same kinetic energy, the de-broglie wavelength is -
 (A) Directly proportional to velocity
 (B) Inversely proportional to velocity
 (C) Inversely proportional to square root of mass
 (D) Inversely proportional to square root of velocity
- Which gas produces brown coloured solution when it is passed through aqueous solution of FeSO_4 ?
 (A) NO (B) CO (C) NH_3 (D) H_2S
- Which of the following compound give ceric ammonium nitrate test ?



- 
 $\xrightarrow[\text{at } 80^\circ\text{C temp}]{\text{Conc. H}_2\text{SO}_4}$ P (major) , product P is?

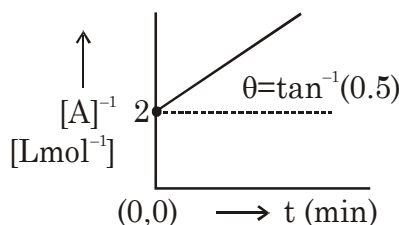


Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
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6. Consider the following graph for the reaction $A \rightarrow P$:

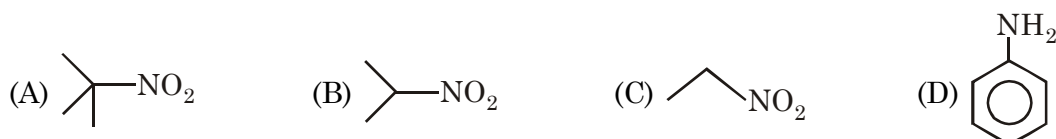


choose the correct statement(s)

- (A) The given reaction is a second order reaction
- (B) The given reaction is a third order reaction
- (C) Initial half life of reaction is 4 min
- (D) Initial concentration of A is 2M

Space for Rough Work

7. Which of the following statements is/are true for critical temperature ?
 (A) It is the highest temperature at which liquid and vapour coexist.
 (B) Above critical temperature, there is no distinction between liquid and gas phases
 (C) At critical temperature, the surface tension of the system is zero
 (D) At critical temperature, the viscosity of the system is zero.
8. In which of the following metal extraction, following steps are involved?
 (i) Roasting (ii) Carbon reduction (iii) Electrolytic refining
 (A) Lead from galena (B) Magnesium from carnelite
 (C) Iron from haematite (D) Zinc from sphalerite
9. Select **CORRECT** statement(s).
 (A) O_3 and O^{2-} both are diamagnetic
 (B) Out of O_2 , O_2^+ and O_2^- , O_2 have maximum magnetic moment (spin only)
 (C) Out of O_2 , O_2^+ and O_3 , O_2^+ have least O-O bond length
 (D) Out of O_2 , O_2^{2+} and O_2^{2-} , only O_2 is paramagnetic.
10. Which of the following produce both Cr_2O_3 and O_2 in product side?
 (A) Reaction of Al_2O_3 with Cr (B) Heating CrO_3
 (C) Heating $(NH_4)_2Cr_2O_7$ (D) Heating $K_2Cr_2O_7$
11. Which of the following is/are reducing sugar
 (A) Sucrose (B) Aldohexose (C) Aldotriose (D) Ketotriose
12. Which of the following compound react with HNO_2 ?



13. Which of the following reaction is/are gives primary alcohol as a major product ?
 (A) $PhMgBr + \triangle \xrightarrow[(2) H_3O^+]{(1) \text{ Dry ether}}$ (B) $PhMgBr \xrightarrow[(2) H_3O^+]{(1) O_2}$
 (C) $PhMgBr + HCHO \xrightarrow[(2) H_3O^+]{(1) \text{ Dry ether}}$ (D) $PhMgBr + CH_3CHO \xrightarrow[(2) H_3O^+]{(1) \text{ Dry ether}}$

Space for Rough Work

SECTION-II : (Maximum Marks : 15)

MATCHING LIST TYPE QUESTION (ANSWER AS TRIPLE DIGIT INTEGER)

- This section contains **FIVE** questions.
- The answer to each question is a **TRIPLE DIGIT INTEGER** ranging from 111 to 333, both inclusive
- This section contains four questions, each having two matching lists. Choices for the correct combination of elements form List-I (P), (Q), (R) and List-II (1), (2), (3)
- Given matching type questions in which list-I (PQR) match with list-II (1, 2, 3) and represent your answer as triplet digit integer for example correct match for $P \rightarrow 2$; $Q \rightarrow 3$; $R \rightarrow 1$ Then you have fill the answer in section-II as 231.

	(+)	(-)
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Each matching in list-II may be/may not be correct/can be used more than one times for matching in list-I. For example your answer can be 223, 333, 133 etc...

- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

1.	List-I(Reaction)	List-II($\Delta_f H$ is known as)
	(P) $C(\text{diamond}) \rightarrow C(g)$	(1) $\Delta_{\text{lattice}} H$
	(Q) $CH_3COOH(aq.) + OH^-(aq.)$	(2) $\Delta_{\text{atomisation}} H$
	↓	
	$CH_3COO^-(aq.) + H_2O(l)$	
	(R) $NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$	(3) $\Delta_{\text{neutralisation}} H$
2.	List-I	List-II
	(P) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(1) $K_p = K_c(RT)^1$
	(Q) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$	(2) $K_c = K_p(RT)^1$
	(R) $2NO_2(g) \rightleftharpoons N_2O_4(g)$	(3) $K_p = K_c$

Space for Rough Work

3. List-I

- (P) Complex follow Sidgwick rule
 (Q) Complex have geometrical isomer
 (R) Complex have optical isomer

List-II

- (1) $[\text{Pt}(\text{en})\text{ClBr}]$
 (2) $\text{Na}[\text{Co}(\text{CO})_4]$
 (3) $[\text{Cr}(\text{en})_2\text{Cl}_2]\text{Cl}$

4. List-I

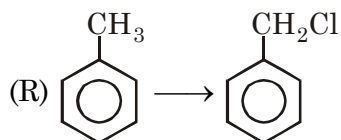
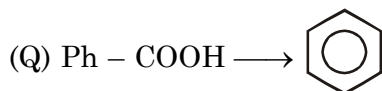
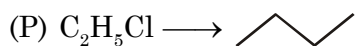
- (P) ZnSO_4 solution
 (Q) CuSO_4 solution
 (R) AgNO_3 solution

List-II

- (1) Gives chocolate brown ppt with $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution
 (2) Gives white ppt with dil HCl
 (3) Gives Reinmann's green with $\text{Co}(\text{NO}_3)_2$ solution followed by heating

5. List-I

Reaction with their major product


List-II

Missing reagent

- (1) NaOH / CaO
 (2) $\text{Na} / \text{Dry ether}$
 (3) $\text{Cl}_2 / h\nu$

Missing reagent for reactions PQR respectively?

Space for Rough Work

PART-3 : MATHEMATICS**SECTION-I(i) : (Maximum Marks : 15)**

- This section contains **FIVE** questions.
 - Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
 - For each question, darken the bubble corresponding to the correct option in the ORS.
 - For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases
-

1. If $f_1(x) = e^x$ and $f_{n+1}(x) = e^{f_n(x)} \forall n \in \mathbb{N}$. Then $\frac{d}{dx}(f_n(x))$ is equal to-
- (A) $f_n(x)$ (B) $f_n(x)f_1(x)$
(C) $f_n(x)f_{n-1}(x)....f_1(x)$ (D) $f_n(x)f_{n-1}(x)....f_1(x)e^x$
2. A normal is drawn at point P of an ellipse, which meets the major axis at point G. If $SP = 2$ and $S'P = 8$, then the ratio of areas of ΔPSG and $\Delta PS'G$ is (where S and S' are foci)
- (A) 4 : 1 (B) 1 : 1 (C) 1 : 2 (D) 1 : 4
-

Space for Rough Work

3. At the beginning of the year 2019, India's population was 130 crore and it is increasing at constant rate of 1% annually. Right now the mobile numbers are of 10 digits starting with digits 6,7,8 or 9 only, as permitted by TRAI (a regulating body). Find the maximum number of years the current pattern of mobile numbers can run, assuming two mobile numbers per member of population (use $\log_{(1.01)}\left(\frac{20}{13}\right) = 43.29$)
- (A) 44 years (B) 43 years (C) 22 years (D) 21 years
4. Two distinct function $f(x)$ and $g(x)$ are chosen randomly from a set of functions $\left\{x, x^2, \sin\left(\frac{1}{x}\right), \cos\left(\frac{1}{x}\right)\right\}$. Then the probability that $\lim_{x \rightarrow 0} f(x)g(x)$ exist, is
- (A) $\frac{5}{6}$ (B) $\frac{1}{12}$ (C) $\frac{1}{3}$ (D) $\frac{1}{6}$
5. If $|z| = 1$, then locus of $w = \frac{z-1}{z+1}$ ($z \neq -1$) is (where z and w are complex numbers)
- (A) a circle (B) imaginary axis (C) real axis (D) a ray

Space for Rough Work

SECTION-I(ii) : (Maximum Marks: 32)

- This section contains **EIGHT** questions.
 - Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
 - For each question, choose the correct option(s) to answer the question.
 - Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -2 In all other cases.
 - **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.
-
6. Choose the correct statement(s)-
- (A) The largest integer in the range of function $(\sin^{-1}x)^2 - (\cos^{-1}x)^2$ is 2
- (B) The smallest integer in the range of function $(\sin^{-1}x)^2 - (\cos^{-1}x)^2$ is -7
- (C) The value of 'x' satisfying $\sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$, is greater than $\frac{1}{2}$
- (D) The value of 'x' satisfying $\sin^{-1}x + 2\cos^{-1}(-x) = 2\pi$, is less than $\frac{1}{2}$
7. If A is a square matrix such that $A^2 = I$, then $\det((A - I)^3 + (A + I)^3 - 7A)$ can be-
 (where $\det(A)$ denotes determinant of square matrix A)
- (A) 0 (B) -1 (C) 1 (D) 2

Space for Rough Work

8. A particle moves over a path, whose position vector at any time t , is given by $\vec{r}(t) = |t|\hat{i} + |\sin \pi t|\hat{j} + \sqrt[3]{t-2}\hat{k}$. A path $\vec{r}(t)$ is called differentiable at time t if its all components are differentiable with respect to 't' at that point of time. So $\vec{r}(t)$ is differentiable at $t =$
 (A) 3 (B) 1 (C) 2 (D) 3.5

9. If $f(x) = \begin{cases} 2\sqrt{x} & ; x \in [0,1) \\ p & ; x = 1 \\ -\ln x & ; x > 1 \end{cases}$. Then at $x = 1$, $f(x)$ has local

(A) maxima if $p = 2$ (B) maxima if $p = 1$ (C) minima if $p = 0$ (D) minima if $p = -1$

10. If $\int e^x (x+1)^2 dx = (x^2 + px + q)e^x + c$ (where c is constant of integration and $p, q \in \mathbb{R}$), then-
 (A) $pq = 0$ (B) $p + q = 1$ (C) $q = 0$ (D) $p = 0$

11. If $\int_0^\infty \frac{x^2 dx}{(x^2 + a^2)(x^2 + b^2)(x^2 + c^2)} = \frac{\pi}{2(a+b)(b+c)(c+a)}$; $a, b, c \in \mathbb{R}$ further, $I = \int_0^\infty \frac{dx}{(x^2 + 4)(x^2 + 9)}$

and $J = \int_0^\infty \frac{dx}{(x^2 + 1)(x^2 + 4)(x^2 + 9)}$, then

(A) $I = \frac{\pi}{60}$ (B) $I = \frac{\pi}{30}$ (C) $J = \frac{\pi}{120}$ (D) $J = \frac{\pi}{240}$

Space for Rough Work

12. Consider the curve

$$S = (\sin\theta)x^2 + 2(\cos\theta)xy + (\sin\theta)y^2 + 2(\cos\theta)x + 2(\cos\theta)y + \sin\theta = 0$$

Now, choose the correct statement(s)

(A) If $\theta = \frac{3\pi}{4}$, then $S = 0$ represents a pair of straight lines.

(B) If $S = 0$ represents a pair of real straight lines, then $\theta = n\pi + \frac{\pi}{4}$, $n \in \mathbb{Z}$

(C) If $S = 0$ represents a pair of real straight lines, then they are parallel.

(D) There exist some θ for which $S = 0$ represents no real locus.

13. Consider a family of curves in which tangent at any point always passes through (1,0). Then

(A) Differential equation governing the given family is $\frac{dy}{dx} = \frac{x-1}{y}$

(B) The general equation of given family is $y = m(x-1)$; m is arbitrary constant

(C) A curve of the given family passing through (2,1) is $y = x - 1$

(D) x-axis belongs to the given family.

Space for Rough Work

SECTION-II : (Maximum Marks : 15)

MATCHING LIST TYPE QUESTION (ANSWER AS TRIPLE DIGIT INTEGER)

- This section contains **FIVE** questions.
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- Given matching type questions in which list-I (PQR) match with list-II (1, 2, 3) and represent your answer as triplet digit integer for example correct match for $P \rightarrow 2$; $Q \rightarrow 3$; $R \rightarrow 1$ Then you have fill the answer in section-II as 231.

	+	-
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Each matching in list-II may be/may not be correct/can be used more than one times for matching in list-I. For example your answer can be 223, 333, 133 etc...

- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.
Zero Marks : 0 In all other cases.

1.	List-I	List-II
(P)	If $x, y > 0$ such that $2x + y = 3$, then maximum value of x^2y is	(1) 1
(Q)	The value of $[-2 \sin 25^\circ]^2$ is equal to (where $[.]$ denotes greatest integer function)	(2) 2
(R)	The value of $\left \frac{\sqrt{2} + \sqrt{3}w + \sqrt{5}w^2}{\sqrt{5} + \sqrt{2}w + \sqrt{3}w^2} \right $ is equal to (w is imaginary cube root of unity)	(3) 4

Space for Rough Work

2. List-I contains vector equation in unknown $\vec{r}$ and List-II contains number of solutions i.e. number of such possible $\vec{r}$.

List-I

List-II

(P) $\vec{r} \times \hat{i} = \hat{i}$

(1) 0

(Q) $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 1$ and $|\vec{r}| = \frac{1}{\sqrt{3}}$

(2) 1

(R) $|\vec{r} - \hat{i}| = 1$

(3) more than 1

3. **List-I**

List-II

(P) The number of roots of equation

(1) 0

$\sin^2 x - 5 \sin x + 6 \leq 0$; $x \in [0, 2\pi]$, is

(Q) The eccentricity of a hyperbola $24x^2 - y^2 = 24$, is

(2) 3

(R) The angle between the tangents drawn from (1,4) to

(3) 5

the parabola $y^2 = 4x$ is $\frac{2\pi}{k}$, where k is equal to

Space for Rough Work

- | 4. | List-I | List-II |
|----|--|---------|
| | (P) If the equation of a plane containing the co-planar lines $\frac{x-2}{1} = \frac{y-9}{2} = \frac{z-a}{3}$ and $\frac{x+3}{1} = \frac{y-1}{-2} = \frac{z+2}{3}$ ($a \in \mathbb{R}$) is $3x - z + k = 0$, then k is equal to | (1) 7 |
| | (Q) AB is a chord of the circle $x^2 + y^2 = 9$. The tangent at A and B intersect at C. If M(1,2) is the mid-point of AB, then the area of $\triangle ABC$ is $\frac{p}{\sqrt{5}}$, where p is equal to | (2) 8 |
| | (R) If number of words that can be formed using all the letters of the word 'ALLEN' is N, then sum of digits of N is | (3) 6 |

Space for Rough Work

5.	List-I	List-II
(P)	If $(1 + x + x^2)^{10} = \sum_{r=0}^{20} a_r x^r$ then $a_5 - \binom{10}{1} a_4 + \binom{10}{2} a_3 - \binom{10}{3} a_2 + \binom{10}{4} a_1 - \binom{10}{5} a_0$ is equal to	(1) 0
(Q)	The period of $f(x) = \frac{e^x}{e^{[x]}}$, is (where $[.]$ denotes greatest integer function)	(2) 5
(R)	The maximum slope of tangent to the curve $y = -x^3 + 3x^2 + 2x$ is	(3) 1

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work